

BUILDING A VIRTUAL FIREWALL IDS AND IPS WITH SURICATA

*A Mini-project Report submitted
in partial fulfilment for the award of the Degree of*

**Bachelor of Technology
in
Computer Science and Engineering
by**

MOHAMMAD SALMAN SALAHUTHEEN (U21CC026)

PULI SUJITH (U21CC035)

SAMEER RAJ (U21CC055)

Under the guidance of

Mrs. D. Bhavana Asst Professor, Dept of CSE



**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
SCHOOL OF COMPUTING
BHARATH INSTITUTE OF HIGHER EDUCATION AND RESEARCH
(Deemed to be University Estd u/s 3 of UGC Act, 1956)
CHENNAI 600 073, TAMIL NADU, INDIA**

December, 2024



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

BONAFIDE CERTIFICATE

This is to Certify that this Mini-Project Report Titled “**BUILDING A VIRTUAL FIREWALL IDS AND IPS WITH SURICATA**” is the Bonafide Work of Mohammad Salman Salahutheen (U21CC026), Puli Sujith (U21CC035), Sameer raj (U21CC055) of Final Year B.Tech. (CSE) who carried out the mini project work under my supervision. Certified further, that to the best of my knowledge the work reported here in does not form part of any other project report or dissertation on basis of which a degree or award conferred on an earlier occasion by any other candidate.

PROJECT GUIDE

Mrs. D.Bhavana

Asst Professor

Department of CSE

BIHER

HEAD OF THE DEPARTMENT

Dr. S. Maruthuperumal

Professor

Department of CSE

BIHER

Submitted for Semester Mini-Project viva-voice examination held on

INTERNAL EXAMINER

EXTERNAL EXAMINER

DECLARATION

We declare that this Mini-project report titled “**BUILDING A VIRTUAL FIREWALL IDS AND IPS WITH SURICATA**” submitted in partial fulfillment of the degree of **B. Tech in (Computer Science and Engineering)** is a record of original work carried out by us under the supervision of **Mrs.D.Bhavana**, and has not formed the basis for the award of any other degree or diploma, in this or any other Institution or University. In keeping with the ethical practice in reporting scientific information, due acknowledgements have been made wherever the findings of others have been cited.

Mohammad Salman Salahutheen
(U21CC026)

Puli Sujith
(U21CC035)

Sameer raj
(U21CC055)

Chennai

Date:

ACKNOWLEDMENTS

We express our heartfelt gratitude to our esteemed Chairman, **Dr.S. Jagathrakshakan, M.P.**, for his unwavering support and continuous encouragement in all our academic endeavors.

We express our deepest gratitude to our beloved President **Dr. J. Sundeep Aanand** President, and Managing Director **Dr. E. Swetha Sundeep Aanand** Managing Director for providing us the necessary facilities to complete our project.

We take great pleasure in expressing sincere thanks to **Dr. K. VijayaBaskar Raju** Pro-Chancellor, **Dr. M. Sundararajan** Vice Chancellor (i/c), **Dr. S. Bhuminathan** Registrar and **Dr. R. Hariprakash** Additional Registrar, **Dr. M. Sundararaj** Dean Academics for moldings our thoughts to complete our project.

We thank our **Dr. S. Neduncheliyan** Dean, School of Computing for his encouragement and the valuable guidance.

We record indebtedness to our Head, **Dr. S. Maruthuperumal**, Department of Computer Science and Engineering for his immense care and encouragement towards us throughout the course of this project.

We also take this opportunity to express a deep sense of gratitude to our Supervisor **Mrs.D.Bhavana** and our Project Co-Ordinator

Dr.D.Selvapriya for their cordial support, valuable information, and guidance, they helped us in completing this project through various stages.

We thank our department faculty, supporting staff and friends for their help and guidance to complete this project.

MOHAMMAD SALMAN SALAHUTHEEN (U21CC026)

PULI SUJITH (U21CC035)

SAMEER RAJ (U21CC055)

Abstract

The increasing prevalence of cyber threats necessitates robust network security solutions that combine detection, prevention, and mitigation of attacks. This project explores the implementation of a virtual firewall integrated with an Intrusion Detection System (IDS) and Intrusion Prevention System (IPS) using Suricata, an open-source network threat detection engine. The objective is to design and configure a virtual security infrastructure capable of identifying, analyzing, and blocking malicious activities in real-time. Suricata is deployed in a virtualized environment to monitor network traffic, leveraging signature-based, anomaly-based, and hybrid detection mechanisms. The virtual firewall is configured to enforce security policies, allowing or blocking traffic based on predefined rules. The IDS component identifies potential threats, while the IPS actively mitigates them, ensuring uninterrupted network performance and enhanced security.

This project outlines the system architecture, configuration of Suricata in conjunction with tools like iptables for traffic control, and the deployment of custom rules to detect DoS attacks, port scans, and other malicious activities. Performance is evaluated by simulating real-world attack scenarios, including Denial of Service (DoS) attacks, and assessing the firewall's response. The results demonstrate Suricata's efficiency in threat detection and its integration into a virtual firewall setup as a cost-effective, scalable solution for modern cybersecurity challenges. This work provides a framework for academic and practical implementations of network security mechanisms using open-source technologies. The project emphasizes the significance of open-source tools in fostering accessibility and innovation in the cybersecurity domain.

Keywords — Virtual Firewall, pfSense, Intrusion Detection System (IDS), Intrusion Prevention System (IPS), Suricata, Network Security, Virtualization, Threat Prevention, Virtual Machine, Threat Detection, blocking features, stateful packet inspection.

LIST OF FIGURES

FIGURE	TITLE	PAGE NUMBER
3.3	Firewall system	20
3.4	Blocked site by firewall	21
3.5	System logs of firewall	21
3.6	Proposed system architecture	23
3.7	LOIC DOS attack application	25
4.4.1	SID Auto-Management List Editor	28
4.4.8	Rule to block DOS attack	30
4.5	Diagram of Block Offenders	32
4.5.6	System logs of IDS	33
5.1	Alerts log	38

Abbreviations/ Notations/ Nomenclature

IDS	Intrusion Detection System
IPS	Intrusion Prevention System
SID	Security Identifier
SIEM	Security Information and Event Management
LOIC	Low Orbit Ion Cannon

TABLE OF CONTENT

DESCRIPTION	PAGENO
ABSTRACT	v
LIST OF FIGURES	vi
ABBREVIATIONS	vii
1 INTRODUCTION	12
1.1 Introduction	12
1.2 Aim of the project	12
1.3 project Domain	13
1.4 Module	13
1.4.1 Virtual Environment Setup.	13
1.4.2 pfsense as a firewall.	13
1.4.3 suricata IDS/IPS integration.	14
1.4.4 Firewall and IDS/IPS Rule Management.	14
1.4.5 Traffic bypass configuration.	14
1.4.6 Logging and Monitoring.	14
1.4.7 Testing and Validation.	15
2 LITERATURE REVIEW	16
3 DESIGN METHODOLOGY	19

3.1	System Requirements	19
3.2	System Design and Planning Architecture Design.	19
3.3	IPS and IDS Integration	20
3.4.	Testing and Validation.	20
3.5.	Deployment.	21
3.6	PROPOSED SYSTEM ARCHITECTURE.	22
3.6.1.	Internet	22
3.6.2.	Router	22
3.6.3.	Firewall.	22
3.6.4.	Switch.	22
3.6.5.	IPS/IDS [Intrusion Prevention/Detection System].	22
3.6.6.	Server.	22
3.7	Attacking Methodology.	23
3.7.1.	Initialization.	23
3.7.2.	Attack Execution.	23
3.7.3.	Modes of Operation.	24
3.7.4.	Target Impact.	24
3.7.5.	Defence and Detection.	24
3.7.6.	Mitigation.	25
4	IMPLEMENTATION	26

4.1. Virtual Environment.	26
4.2. Network Configuration.	26
4.3. Example Setup in VirtualBox.	27
4.4 blocking dos attack in ids.	27
4.4.1. Enable IPS Mode in Suricata.	27
4.4.2. Update Rules.	28
4.4.3. Enable Relevant Rules for DoS Detection.	28
4.4.4. Tune Rules for Specific DoS Patterns.	28
4.4.5. Adjust Performance Settings.	29
4.4.6. Monitor and Test.	29
4.4.7. Add Custom Rules if Needed.	29
4.4.8. Enable External Protection.	29
4.5 blocking attack in ips.	30
4.5.1. Setup Overview.	30
4.5.2. Configure Suricata in IPS Mode.	30
4.5.3. Simulate the DoS Attack.	31
4.5.4. Detect and Block with Suricata.	31
4.5.5. Fine-Tune the Protection.	32
4.5.6. Validate the Protection.	33
4.6 Suricata installation.	34

4.7pfsense firewall installation.	35
5 RESULTS AND DISCUSSION	38
5.1 Results.	38
5.2 Discussion.	39
6 CONCLUSION AND FUTURE SCOPE	40
6.1 Conclusion.	40
6.2 Future scope.	40
REFERENCES	41

CHAPTER 1

INTRODUCTION

1.1 Introduction

In today's interconnected world, organizations face a relentless onslaught of cyber threats. These attacks, ranging from simple phishing to sophisticated ransomware campaigns, can disrupt operations, inflict financial losses, and damage reputation.

To safeguard against these threats, a robust security infrastructure is imperative. This project aims to enhance network security by deploying a virtual firewall fortified with intrusion detection and prevention capabilities.

By integrating pfSense with Suricata, we seek to create a resilient defense against the evolving threat landscape and provide organizations with a cost-effective yet powerful security solution. pfSense, an open-source firewall and router platform, offers extensive customization and scalability, making it an ideal choice for businesses of all sizes. By integrating Suricata, a high-performance network intrusion detection and prevention system (IDS/IPS), this project will demonstrate how advanced threat detection and proactive defense mechanisms can be implemented seamlessly.

The integration of pfSense and Suricata allows for deep packet inspection, real-time threat analysis, and automated response to malicious activities. The solution leverages open-source technology, reducing dependency on expensive proprietary systems while maintaining robust protection against cyber threats.

1.2 Aim of the project

As organizations increasingly rely on digital infrastructure, they face a growing threat from cyber-attacks such as unauthorized access, malware, and denial-of-service (DoS) attacks. Protecting sensitive data and ensuring the availability of critical services have become paramount in today's connected world.

This project focuses on building a virtual firewall with Intrusion Detection System (IDS) and Intrusion Prevention System (IPS) capabilities using pfSense and Suricata.

The goal of this project is to demonstrate how this virtual firewall can be configured to provide comprehensive network protection by combining firewall rules and IDS/IPS functionality.

1.3 Project Domain

The domain for the project "Building a Virtual Firewall IDS and IPS with Suricata" is Cybersecurity, specifically focusing on Network Security within this broader domain. Cybersecurity is the practice of protecting digital assets, systems, and networks from cyber threats, ensuring the confidentiality, integrity, and availability of data. Network security, a crucial pillar of cybersecurity, involves safeguarding communication channels, preventing unauthorized access, and defending against malicious activities that target networked environments.

This project contributes to the cybersecurity domain by addressing the increasing need for robust, scalable, and cost-effective solutions to mitigate cyber threats. It focuses on implementing a virtual firewall fortified with Intrusion Detection and Prevention Systems (IDS/IPS) to monitor and defend against malicious activities in real-time. By leveraging technologies like pfSense and Suricata, the project not only enhances the security infrastructure but also aligns with the goals of cybersecurity to ensure resilient and secure network operations.

1.4 Module:

1.4.1 Virtual Environment Setup:

Objective: To create a virtualized network infrastructure to host the firewall and IDS/IPS.

Description: This module involves setting up a virtual environment using hypervisor VirtualBox. It includes creating virtual machines and configuring network interfaces (WAN, LAN) that will simulate real-world network traffic.

1.4.2. pfSense as a Firewall:

Objective: To install and configure pfSense as the primary firewall for network traffic management.

Description: This module involves installing pfSense on a virtual machine and configuring it to act as the firewall. Firewall rules will be set to control network traffic, allowing or blocking traffic based on IP addresses, ports, and protocols.

1.4.3. Suricata IDS/IPS Integration:

Objective: To install and configure Suricata to provide IDS and IPS functionality within pfSense.

Description: This module covers the installation of the Suricata package in pfSense, configuring it to monitor network traffic, and setting up rule sets for intrusion detection and prevention. This includes enabling either IDS (Intrusion Detection System) for alerting or IPS (Intrusion Prevention System) for actively blocking malicious traffic.

1.4.4. Firewall and IDS/IPS Rule Management:

Objective: To configure firewall rules and IDS/IPS signatures for effective traffic filtering and threat detection.

Description: This module focuses on creating and managing rules in both pfSense and Suricata. In pfSense, you will define firewall rules to control traffic flow between the WAN and LAN. In Suricata, you will configure rule sets to detect and prevent various attacks, such as DDoS, malware, and port scans. The goal is to ensure only authorized traffic is allowed while malicious traffic is blocked.

1.4.5. Traffic Bypass Configuration:

Objective: To demonstrate the “Bypass Firewall Rules for Traffic on the Same Interface” setting and its effects.

Description: This module focuses on configuring pfSense to bypass firewall rules for traffic originating and terminating on the same interface. It includes testing this configuration and observing how trusted internal traffic can bypass firewall scrutiny, ensuring efficient internal communication without compromising security.

1.4.6. Logging and Monitoring:

Objective: To enable logging and monitor security events in both pfSense and Suricata.

Description: This module focuses on configuring pfSense to bypass firewall rules for traffic originating and terminating on the same interface. It includes testing this

configuration and observing how trusted internal traffic can bypass firewall scrutiny, ensuring efficient internal communication without compromising security.

1.4.7. Testing and Validation:

Objective: To test the firewall and IDS/IPS functionality by simulating network traffic and attacks.

Description: This module involves testing the overall system by generating normal and malicious network traffic, such as simulating DDoS, port scanning, and other attacks. The goal is to validate that legitimate traffic passes through while the firewall and IDS/IPS effectively block or alert on malicious activity. This ensures the system's accuracy and performance.

CHAPTER 2

LITERATURE REVIEW

Author: Albin and Rowel (2012)

This paper conducts a comparative analysis of Suricata and Snort, two widely used open-source Intrusion Detection Systems (IDS). The focus is on their performance in real-world network environments characterized by modern and evolving cyber threats. The study highlights Suricata's ability to utilize multithreading, which significantly improves its packet analysis speed and enables it to handle higher traffic loads compared to Snort. Additionally, Suricata's capability to process multiple rules simultaneously makes it more effective in detecting complex attack patterns. The research concludes that while Snort is resource-efficient for smaller networks, Suricata is better suited for high-performance environments requiring scalability and robustness.

Author: Alhasan and Surantha (2021)

This research explores the deployment of next-generation firewalls in data centers, integrating Suricata for enhanced security. The study emphasizes the need for scalable and robust security solutions in large-scale network environments, where traditional firewalls may fall short. Suricata is employed to complement firewall capabilities, offering deep packet inspection, signature-based detection, and anomaly detection. The authors highlight the benefits of using Suricata for detecting complex multi-vector attacks that often target data centers. They also discuss the role of policy management and automated threat response in improving the system's efficiency. This integration demonstrates a significant improvement in the overall security and reliability of data center networks.

Authors: Anggoro and Sulisty (2019), Anugrah et al.(2022)

Both studies explore the use of Suricata for preventing SQL injection attacks, a common and severe threat to web applications. The authors configured Suricata with custom rules to detect SQL injection patterns within HTTP requests and database queries. The system demonstrated high accuracy in identifying malicious queries while maintaining low false positive rates. The

2022 study further extends the analysis by integrating Suricata with additional monitoring tools to enhance detection capabilities. Both papers emphasize Suricata's suitability for securing web applications and highlight its ease of customization to meet specific security needs. These findings are particularly relevant for organizations aiming to protect sensitive data from database breaches.

Author: Praptodiyono et al. (2023)

This paper introduces a hybrid intrusion detection and prevention system by integrating Suricata with pfSense, focusing on IPv6 networks. The research addresses the increasing prevalence of DDoS attacks targeting IPv6 protocols, which pose unique challenges due to their complex addressing mechanisms. The authors designed a system that leverages Suricata's powerful rule-based detection to monitor traffic and pfSense's firewall capabilities to block malicious packets. Testing revealed that this hybrid approach could significantly reduce the impact of DDoS attacks, improving the overall security posture of IPv6-enabled networks. The study emphasizes the importance of adapting IDS/IPS systems to evolving network architectures.

Author: Setiawan and Sulistyo (2023)

This paper presents a novel SIEM (Security Information and Event Management) model combining Suricata with EveBox to detect and respond to malware attacks. Suricata serves as the primary detection engine, monitoring network traffic and generating logs for suspicious activities, while EveBox provides a centralized platform for analyzing and visualizing these events. The integration enables real-time monitoring, detection, and response to malware threats, reducing incident response times. The study showcases the system's effectiveness through simulated malware attacks, highlighting its capability to provide actionable insights. The authors suggest that this model is particularly useful for organizations requiring centralized threat management with minimal resource overhead.

Author: Syani (2020)

In this research, Suricata is deployed on a Linux Debian 9 cloud-based Virtual Private Server (VPS) to evaluate its effectiveness in virtualized environments. The study demonstrates Suricata's adaptability to cloud-based infrastructures, where resource efficiency and scalability are critical. The author configured Suricata to monitor network traffic for potential threats while

maintaining low latency. Testing revealed that Suricata could effectively detect and prevent common attacks, such as port scanning and unauthorized access attempts, in a virtualized setting. The research underscores the importance of using lightweight and flexible IDS solutions like Suricata in cloud-based networks.

Author: Amir Ali and Muhammad Murtaza Yousaf (2020)

Intrusion Detection and Prevention Systems (IDPS) are vital components in securing network environments against malicious activities. Traditional IDPS solutions were developed for static, hardware-based architectures, which often lacked the flexibility and scalability required to handle the dynamic nature of modern networks. The advent of Software-Defined Networking (SDN) has introduced new paradigms in network management by decoupling the control and data planes, enabling centralized network control, programmability, and enhanced scalability. However, the centralized architecture of SDN presents unique security challenges, such as potential single points of failure and increased vulnerability to sophisticated attacks. Researchers have explored various approaches to integrate IDPS with SDN to address these challenges, leveraging SDN's programmability to enhance real-time detection and mitigation capabilities.

CHAPTER 3

DESIGN METHODOLOGY

This design methodology provides a comprehensive approach to building a virtual firewall with IDS/IPS capabilities using Suricata.

3.1. System Requirements:

Hardware:

CPU: 4 cores or more

Memory: 8GB RAM or more

Storage: 50GB SSD or more

Network Interface: Two network interfaces (one for management, one for traffic inspection)

Software:

Operating System: Linux (Ubuntu, Debian, CentOS, etc.)

Suricata: IDS/IPS engine

3.2. System Design and Planning

Architecture Design:

Plan a virtualized environment using platforms like VMware, VirtualBox, or Proxmox.

Map the network topology, including interfaces for external (WAN), internal (LAN), and optional DMZ connections.

Resource Allocation:

Determine system requirements for the virtual machines (CPU, memory, storage, and network bandwidth).

3.3 IPS and IDS Integration:

Configure Suricata as an Intrusion Detection System (IDS) to monitor network traffic for suspicious activity.

Enable Intrusion Prevention System (IPS) mode to automatically block malicious traffic.

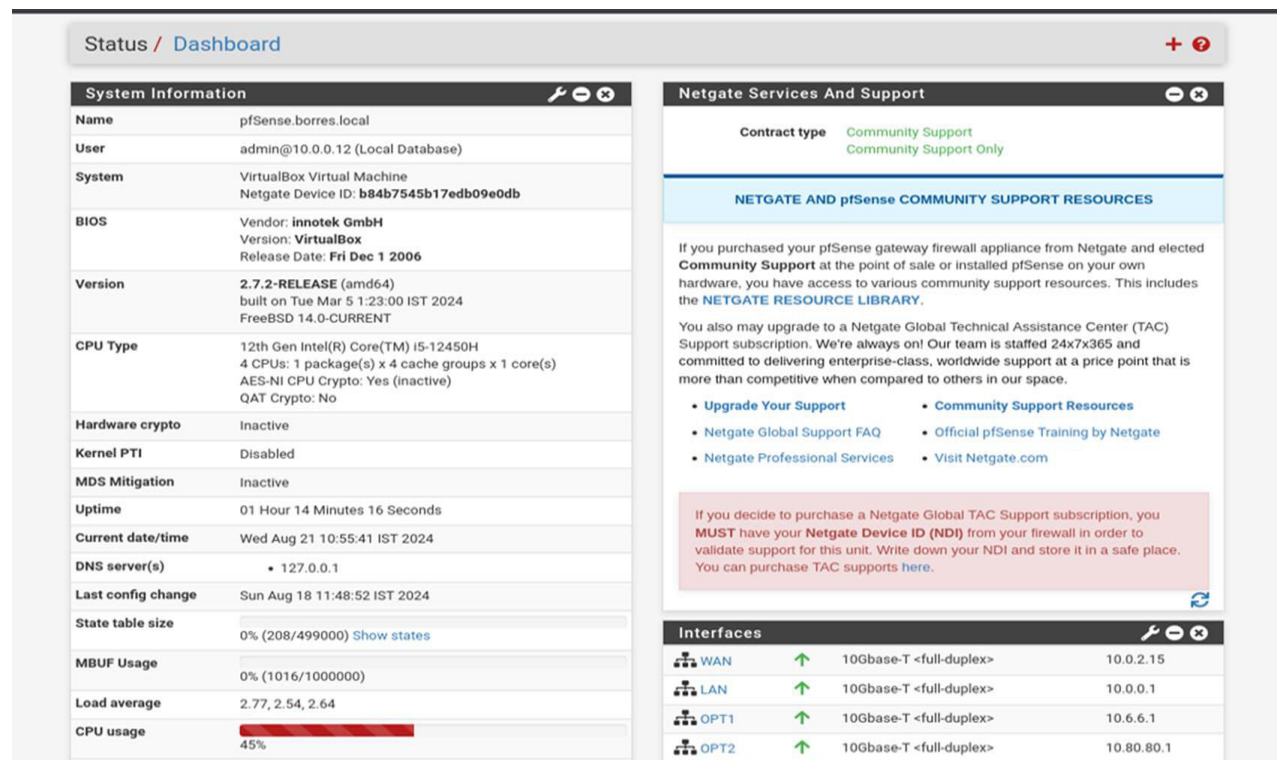


Fig: 3.3 firewall system

3.4. Testing and Validation

Simulation of Traffic:

Generate normal and malicious traffic to test detection and prevention capabilities.

Log Analysis:

Use pfSense and Suricata logs to verify detection and response to simulated threats.

Rule Optimization:

Fine-tune rules to minimize false positives and improve detection accuracy.

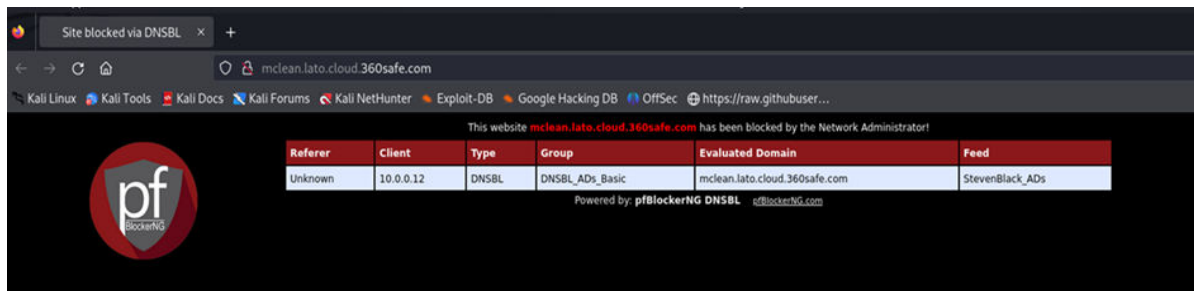


Fig: 3.4 Blocked site by firewall

3.5. Deployment

Production Environment Setup:

Deploy the system in a real-world scenario, ensuring minimal disruption to existing services.

Monitoring Tools:

Set up dashboards and alerting systems for real-time monitoring.

Status / System Logs / Firewall / Normal View						
System Firewall DHCP Authentication IPsec PPP PPPoE/L2TP Server OpenVPN NTP Packages Settings						
Normal View Dynamic View Summary View						
Last 139 Firewall Log Entries. (Maximum 500)						
Action	Time	Interface	Rule	Source	Destination	Protocol
✗	Aug 3 21:02:51	WAN	Default deny rule IPv4 (1000000103)	10.0.2.2:53	10.0.2.15:19742	TCP:RA
✗	Aug 3 21:29:15	LAN	Default deny rule IPv4 (1000000104)	10.0.0.2:60892	10.0.0.1:443	TCP:PA
✗	Aug 3 21:29:15	LAN	Default deny rule IPv4 (1000000104)	10.0.0.2:60892	10.0.0.1:443	TCP:PA
✗	Aug 3 21:29:15	LAN	Default deny rule IPv4 (1000000104)	10.0.0.2:60892	10.0.0.1:443	TCP:PA
✗	Aug 3 21:29:16	LAN	Default deny rule IPv4 (1000000104)	10.0.0.2:60892	10.0.0.1:443	TCP:PA
✗	Aug 3 21:29:16	LAN	Default deny rule IPv4 (1000000104)	10.0.0.2:60892	10.0.0.1:443	TCP:PA
✗	Aug 3 21:29:17	LAN	Default deny rule IPv4 (1000000104)	10.0.0.2:60892	10.0.0.1:443	TCP:PA
✗	Aug 3 21:29:19	LAN	Default deny rule IPv4 (1000000104)	10.0.0.2:60892	10.0.0.1:443	TCP:PA
✗	Aug 3 21:29:19	lo0	(1000002520)	10.0.0.1:443	10.0.0.2:60892	TCP:A
✗	Aug 3 21:29:19	lo0	(1000002520)	10.0.0.1:443	10.0.0.2:60892	TCP:A
✗	Aug 3 21:29:19	lo0	(1000002520)	10.0.0.1:443	10.0.0.2:60892	TCP:A
✗	Aug 3 21:29:19	lo0	(1000002520)	10.0.0.1:443	10.0.0.2:60892	TCP:A
✗	Aug 3 21:29:19	lo0	(1000002520)	10.0.0.1:443	10.0.0.2:60892	TCP:A

Fig: 3.5 System logs of firewall

3.6 PROPOSED SYSTEM ARCHITECTURE:

3.6.1.INTERNET:

Represents the external world or wide area network (WAN). This is where traffic originates or is sent to, such as browsing websites, downloading files, or accessing cloud services.

3.6.2.Router:

The router connects the internal network (LAN) to the external network (internet). It directs traffic between these two networks, ensuring that data reaches the correct destination, either within the local network or out on the internet.

3.6.3.Firewall:

The firewall is positioned between the router and the internal network. It filters incoming and outgoing traffic, blocking unauthorized access while allowing legitimate traffic through. It enforces security rules to protect the network.

3.6.4. Switch:

A network switch connects multiple devices (such as servers, computers, and network devices) within the internal network. It directs data to the correct destination within the local area network (LAN).

3.6.5. IPS/IDS (Intrusion Prevention/Detection System):

This device monitors network traffic for signs of suspicious activity or attacks. The IDS (Intrusion Detection System) detects potential threats, while the IPS (Intrusion Prevention System) actively blocks these threats to protect the network.

3.6.6. Server:

This represents one or more servers that are part of the internal network. Servers could be hosting websites, databases, or other applications that need protection from external threats.

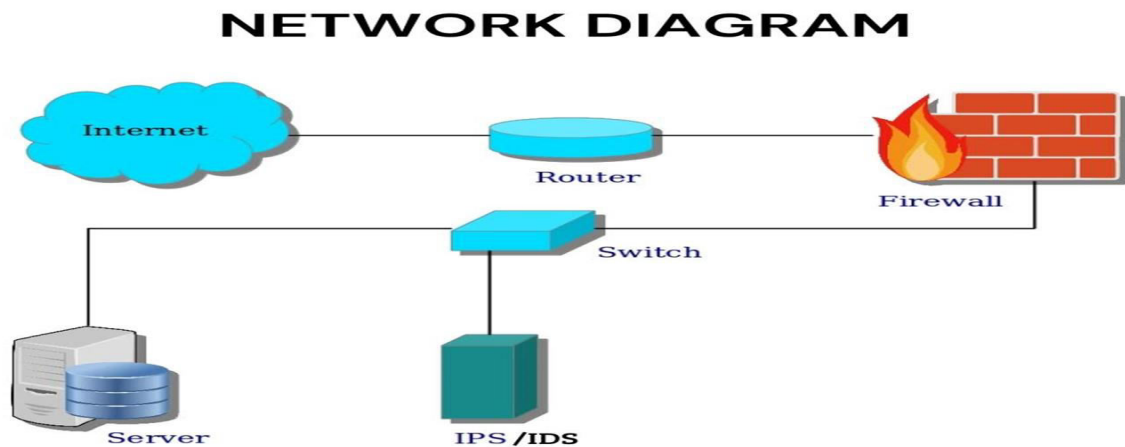


Fig: 3.6 PROPOSED SYSTEM ARCHITECTURE

3.7 Attacking Methodology

The Low Orbit Ion Cannon (LOIC) is an open-source network stress testing and denial-of-service (DoS) attack tool. Understanding its methodology can help in studying and preventing potential DoS attacks. Here's an overview of the LOIC methodology:

3.7.1. Initialization

Target Specification:

The attacker inputs the target's IP address, domain name, or URL.

Ports (e.g., HTTP, HTTPS, or custom) are specified for attack focus.

Attack Method Selection:

LOIC allows users to choose attack types like TCP, UDP, or HTTP requests.

3.7.2. Attack Execution

Payload Generation:

LOIC generates a large number of packets or HTTP requests to the target.

Payloads may contain randomized data or specific patterns.

Flooding Technique:

Continuous high-volume traffic is sent to overwhelm the target.

LOIC uses a straightforward, brute-force flooding approach without stealth or obfuscation.

3.7.3. Modes of Operation

Single-User Mode:

One user uses LOIC to attack a target, causing limited disruption unless the target is small or unprotected.

Hive Mode (Voluntary Botnet):

Users join forces via IRC or other communication platforms to coordinate attacks. This is often referred to as "Anonymous mode," popularized during hacktivist campaigns.

3.7.4. Target Impact

Bandwidth Saturation:

High-volume requests consume the target's bandwidth.

Resource Exhaustion:

Targets' servers struggle to process the volume of incoming requests, causing service degradation or downtime.

Connection Overload:

TCP or UDP floods can overwhelm the server's connection handling capacity.

3.7.5. Defence and Detection

Firewalls and IDS/IPS:

Set up rate-limiting rules and intrusion detection to identify and block unusual traffic patterns.

Traffic Filtering:

Drop packets from IP addresses involved in LOIC-based attacks.

Load Balancers:

Use load balancers to distribute traffic and mitigate DoS attacks.

3.7.6. Mitigation

Approaches in Practice

If you're working with Suricata on pfSense, here's how you can detect and block LOIC activity:

Suricata Rule Configuration:

Use rules to detect abnormal traffic patterns, e.g., high rates of identical requests or unusual UDP/TCP traffic.

Block Offenders:

Under *Suricata > Interface Settings*, enable the Block Offenders option to drop packets from offending IPs.

Monitor Traffic:

Analyse network traffic using tools like Wireshark to identify patterns unique to LOIC.

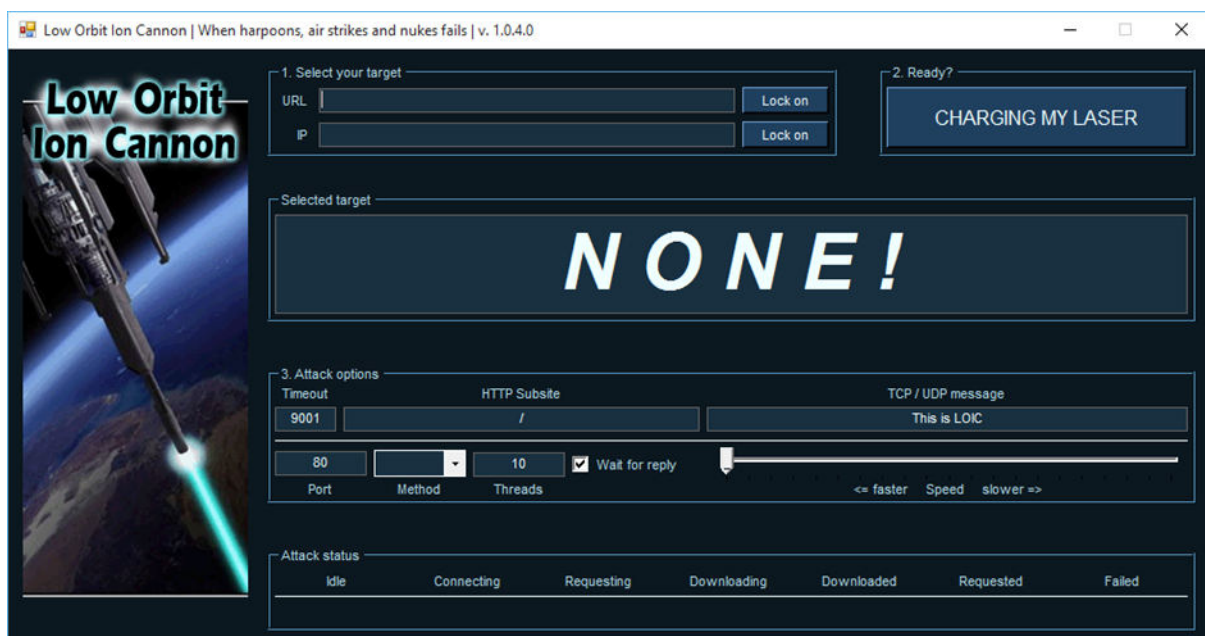


Fig: 3.7 LOIC DOS attack application

CHAPTER 4

IMPLEMENTATION

Infrastructure Setup

4.1. Virtual Environment

To simulate a real-world network securely, you'll need virtualization software. Popular options include:

VirtualBox: Free and open-source virtualization tool.

You will create three types of virtual machines (VMs):

pfSense (Firewall): Acts as the central virtual firewall with Suricata installed.

Client (User): Simulates a regular user generating legitimate traffic. (e.g., browsing, video streaming, etc.)

Attacker: Simulates a malicious entity launching attacks (e.g., Nmap scans, DoS, or Metasploit exploits).

Each VM requires appropriate resources:

pfSense VM: Minimum 4 CPU, 4 GB RAM, 10 GB disk space.

Client/Attacker VMs: Minimum 6 CPU, 5 GB RAM, 50 GB disk space.

4.2. Network Configuration

A proper virtual network design is critical to ensuring traffic flows correctly and can be analyzed:

WAN Interface: Represents the external network (internet).

Assign this to your host machine's real network adapter or use NAT for outgoing connectivity.

This is where external threats can originate.

LAN Interface: Represents the internal (private) network.

Set up an isolated virtual network adapter for communication between the pfSense firewall and the Client VM. This is where normal user traffic will be simulated.

DMZ Interface (Optional): Used to host public-facing services like web servers.

Add this as a separate virtual network for more advanced setups.

You can use this to simulate hosting services while still protecting sensitive internal systems.

4.3. Example Setup in VirtualBox

pfSense VM: Add three network adapters:

Adapter 1: Bridged/NAT (WAN).

Adapter 2: Internal Network (LAN).

Adapter 3: Internal Network (DMZ, optional).

Client VM:

Network Adapter: Internal Network (LAN).

Attacker VM: Network Adapter: Same network as pfSense's WAN.

4.4 blocking dos attack in ids:

Blocking a DoS (Denial of Service) attack in an IDS (Intrusion Detection System) like Suricata involves using its Intrusion Prevention System (IPS) capabilities.

4.4.1. Enable IPS Mode in Suricata

Navigate to Services > Suricata > Interfaces.

Select the interface where you want to monitor traffic.

Under the General Settings tab, enable the Block Offenders option. This will allow Suricata to drop packets that match DoS rules.

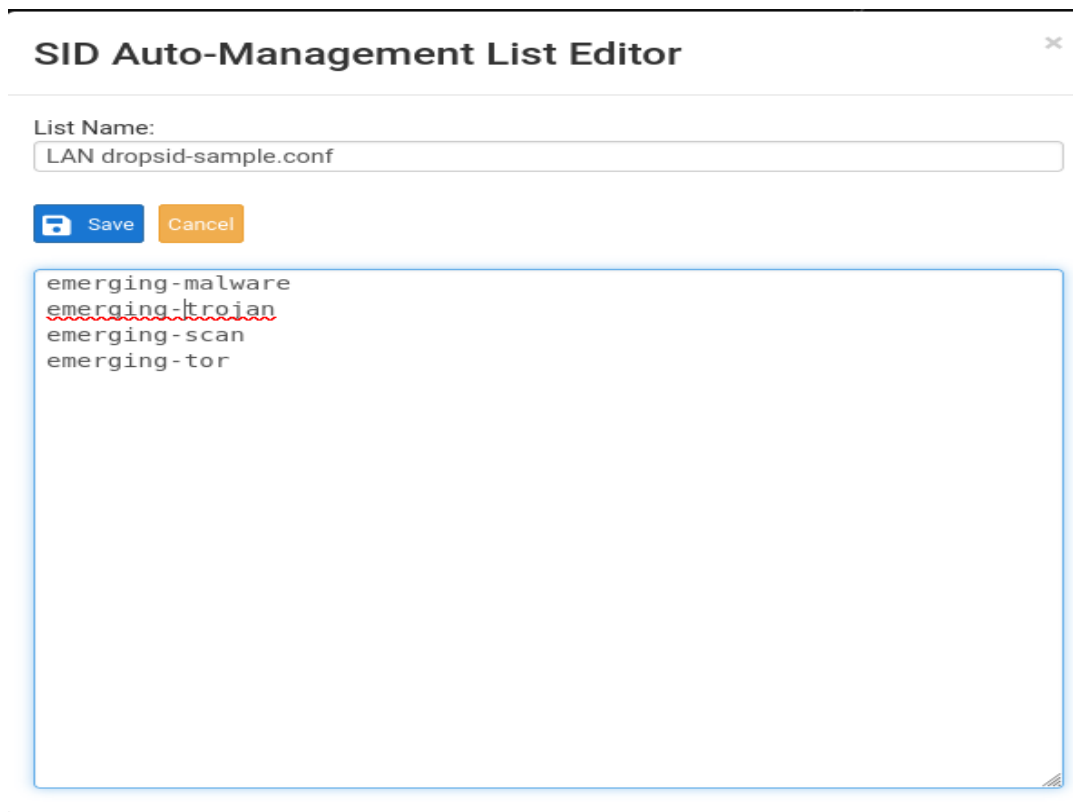


Fig: 4.4.1 SID Auto-Management List Editor

4.4.2. Update Rules

Go to Services > Suricata > Global Settings.

Ensure Rules Update Settings is configured correctly, and that you have enabled relevant rule sources like ET Open Rules or other premium sources if available.

Click Update Rules to ensure the latest signatures are available.

4.4.3. Enable Relevant Rules for DoS Detection

Go to Services > Suricata > Rules.

Search for rules related to DoS attacks. You can filter for keywords like DOS, DDoS, or Denial of Service.

Enable these rules to ensure they trigger alerts or block offending traffic.

4.4.4. Tune Rules for Specific DoS Patterns

For some environments, generic rules may trigger false positives. Fine-tune:

Thresholds: Adjust the number of events needed to trigger an alert or block.

Protocol-specific rules: Focus on the protocols most used in your environment (e.g., HTTP, DNS, ICMP).

4.4.5. Adjust Performance Settings

DoS attacks generate a high volume of traffic, which can overwhelm IDS systems.

Optimize Suricata's performance:

Enable multi-threading in Suricata settings.

Increase the memory and CPU resources allocated to pfSense and Suricata.

4.4.6. Monitor and Test

Use tools like hping3, LOIC, or Metasploit in a controlled environment to simulate

DoS attacks and verify Suricata's detection and blocking.

Monitor alerts under Status > System Logs > Suricata Alerts.

4.4.7. Add Custom Rules if Needed

If specific patterns of DoS are not detected by existing rules, you can create custom Suricata rules:

Replace count and seconds with thresholds suitable for your network.

4.4.8. Enable External Protection

While Suricata can block DoS locally, consider implementing other protection layers, such as:

Rate-limiting in pfSense firewall rules.

Anti-DoS features on upstream routers or services.

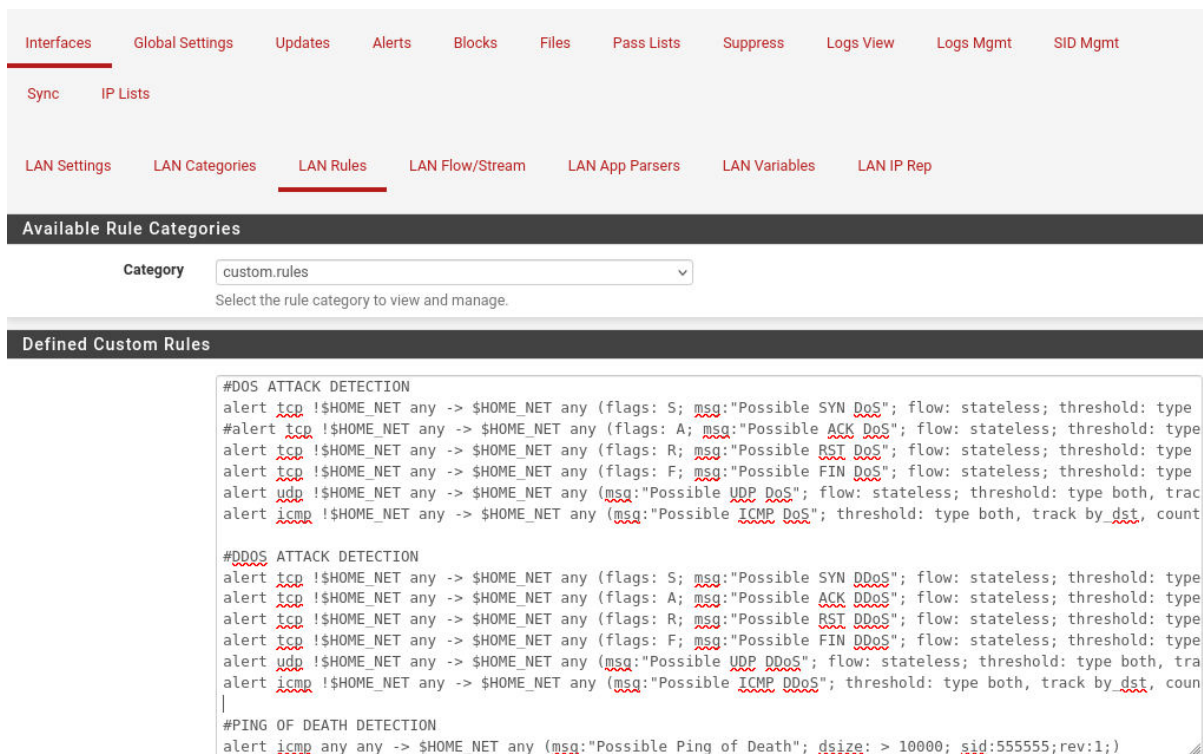


Fig: 4.4.8 Rule to block dos attack

4.5 blocking attack in ips

Blocking a DoS attack from a tool like LOIC (Low Orbit Ion Cannon) using Suricata in IPS mode is a valuable demonstration of how an Intrusion Prevention System (IPS) can protect a network from high-volume traffic attacks.

4.5.1. Setup Overview

Environment: pfSense with Suricata configured.

Attack Simulation: LOIC generating DoS traffic.

Protection: Suricata in IPS mode to detect and block malicious traffic.

4.5.2. Configure Suricata in IPS Mode

Enable IPS Mode on Suricata:

Go to Services > Suricata > Interfaces.

Select the interface where traffic will be monitored (e.g., LAN, WAN).

Enable **IPS Mode** by checking **Block Offenders**.

Update Suricata Rules:

Go to Services > Suricata > Global Settings.

Ensure rule sources like **ET Open Rules** are enabled.

Click **Update Rules** to download the latest detection patterns.

Select Relevant Rules for DoS Protection:

Go to Services > Suricata > Interfaces > Categories.

Enable rule categories related to DoS/DDoS, such as:

dos.rules

emerging-dos.rules

Save and apply the settings.

4.5.3. Simulate the DoS Attack

Set up LOIC on an Attacker Machine:

Use a separate machine or VM to install and run LOIC.

Enter the IP address of your test target (e.g., a web server behind pfSense).

Start the attack by generating HTTP, UDP, or TCP flood traffic.

Monitor the Attack Traffic:

On pfSense, go to Diagnostics > States to see the connections generated by LOIC.

Verify the high volume of traffic attempting to overwhelm the target.

4.5.4. Detect and Block with Suricata

Enable Logging:

Go to Services > Suricata > Interfaces > Logging Settings.

Ensure logging is enabled for alerts and blocked traffic.

Review Alerts:

After starting the attack, go to Services > Suricata > Alerts.

Look for signatures triggered by LOIC traffic, such as high connection rates or malformed packets.

Enable Auto-Blocking:

Suricata automatically blocks IPs when **IPS mode** and **Block Offenders** are enabled. Blocked IPs can be viewed under Diagnostics > pfBlockerNG or Firewall > Aliases.

Firewall / pfBlockerNG

GeneralIPDNSBLUpdateReportsFeedsLogsSyncWizard

General Settings

LinksFirewall AliasesFirewall RulesFirewall Logs

pfBlockerNG

☒ Enable

Note: Context help is available on various pages by clicking the 'blue infoblock' icons

Keep Settings

☒ Enable

Note: With 'Keep settings' enabled, pfBlockerNG will maintain run state on Installation/Upgrade. If 'Keep Settings' is not 'enabled' on pkg Install/De-Install, all settings will be Wiped!

Note: To clear all downloaded lists, uncheck these two checkboxes and 'Save'. Re-check both boxes and run a 'Force Update|Reload'

CRON Settings

Every hour0000

Default: Every hour
Select the Cron hour interval.

Default: :00
Select the Cron update minute.

Default: 0
Select the Cron start hour.

Default: 0
Select the 'Daily/Weekly' start hour.

Download Failure Threshold

No Limit

Default: No limit
Select max daily download failure threshold via CRON. Clear widget 'failed downloads' to reset. On a download failure, the previously downloaded list is reloaded.

Log Settings (max lines)

General

20,00020,00020,00020,000

Default: 20000
pfBlockerNG Log

Default: 20000
Unified Log Log

Default: 20000
Error Log

Default: 20000
Extras Log

IP

20,00020,00020,000

Default: 20000
IP Block Log

Default: 20000
IP Permit Log

Default: 20000
IP Match Log

Fig: 4.5 Diagram of Block Offenders

4.5.5. Fine-Tune the Protection

Adjust Sensitivity:

Modify rule thresholds to detect excessive connections more quickly.
Add custom rules for specific patterns if LOIC's traffic isn't being caught effectively.

Log and Analyze:

Use Status > System Logs > Firewall to verify blocked connections.
Export Suricata logs to analyze attack patterns and refine your defenses.

4.5.6. Validate the Protection

Rerun the Attack:

Start LOIC again and observe how Suricata blocks the traffic in real time.

Check the Logs:

Confirm that Suricata has successfully identified and blocked the attack without affecting legitimate traffic.

Test for False Positives:

Ensure the rules don't inadvertently block non-malicious traffic.

Status / System Logs / System / General											
System	Firewall	DHCP	Authentication	IPsec	PPP	PPPoE/L2TP Server	OpenVPN	NTP	Packages	Settings	
General	Gateways	Routing	DNS Resolver	Wireless	GUI Service	OS Boot					
Last 500 General Log Entries. (Maximum 500)											
Time	Process	PID	Message								
Nov 19 19:42:47	suricata	27304	{\"timestamp\":\"2024-11-19T19:42:47.252882+0530\",\"flow_id\":\"2045063968051776\",\"in_iface\":\"vtnet0\",\"event_type\":\"dns\",\"src_ip\":\"10.0.2.15\",\"src_port\":23726,\"dest_ip\":\"198.97.190.53\",\"dest_port\":53,\"proto\":\"UDP\",\"pkt_src\":\"wire/pcap\",\"dns\":{\"version\":2,\"type\":\"answer\",\"id\":5871,\"flags\":\"8400\",\"qr\":true,\"aa\":true,\"opcode\":0,\"rname\":\"\",\"rrtype\":\"NS\",\"rcode\":\"NOERROR\",\"answers\":{\"rname\":\"\",\"rrtype\":\"NS\",\"ttl\":518400,\"rdata\":\"a.root-servers.net\"},{\"rname\":\"\",\"rrtype\":\"NS\",\"ttl\":518400,\"rdata\":\"b.root-servers.net\"},{\"rname\":\"\",\"rrtype\":\"NS\",\"ttl\":518400,\"rdata\":\"c.root-servers.net\"},{\"rname\":\"\",\"rrtype\":\"NS\",\"ttl\":518400,\"rdata\":\"d.root-servers.net\"},{\"rname\":\"\",\"rrtype\":\"NS\",\"ttl\":518400,\"rdata\":\"e.root-servers.net\"},{\"rname\":\"\",\"rrtype\":\"NS\",\"ttl\":518400,\"rdata\":\"f.root-servers.net\"},{\"rname\":\"\",\"rrtype\":\"NS\",\"ttl\":518400,\"rdata\":\"g.root-servers.net\"},{\"rname\":\"\",\"rrtype\":\"NS\",\"ttl\":518400,\"rdata\":\"h.root-servers.net\"},{\"rname\":\"\",\"rrtype\":\"NS\",\"ttl\":518400,\"rdata\":\"i.root-servers.net\"},{\"rname\":\"\",\"rrtype\":\"NS\",\"ttl\":518400,\"rdata\":\"j.root-servers.net\"},{\"rname\":\"\",\"rrtype\":\"NS\",\"ttl\":518400,\"rdata\":\"k.root-servers.net\"},{\"rname\":\"\",\"rrtype\":\"NS\",\"ttl\":518400,\"rdata\":\"l.root-servers.net\"},{\"rname\":\"\",\"rrtype\":\"NS\",\"ttl\":518400,\"rdata\":\"m.root-servers.net\"},{\"rname\":\"\",\"rrtype\":\"RRSIG\",\"ttl\":518400,\"grouped\":{\"NS\":{\"a.root-servers.net\",\"b.root-servers.net\",\"c.root-servers.net\",\"d.root-servers.net\",\"e.root-servers.net\",\"f.root-servers.net\",\"g.root-servers.net\",\"h.root-servers.net\",\"i.root-servers.net\",\"j.root-servers.net\",\"k.root-servers.net\",\"l.root-servers.net\",\"m.root-servers.net\"}}}}								
Nov 19 19:42:47	suricata	27304	{\"timestamp\":\"2024-11-19T19:42:47.255022+0530\",\"flow_id\":\"2221214237253448\",\"in_iface\":\"vtnet0\",\"event_type\":\"dns\",\"src_ip\":\"10.0.2.15\",\"src_port\":40545,\"dest_ip\":\"192.5.5.241\",\"dest_port\":53,\"proto\":\"UDP\",\"pkt_src\":\"wire/pcap\",\"dns\":{\"type\":\"query\",\"id\":24894,\"rname\":\"com\",\"rrtype\":\"A\",\"tx_id\":0,\"opcode\":0}}								
Nov 19 19:42:47	suricata	27304	{\"timestamp\":\"2024-11-19T19:42:47.270291+0530\",\"flow_id\":\"2221214237253448\",\"in_iface\":\"vtnet0\",\"event_type\":\"dns\",\"src_ip\":\"10.0.2.15\",\"src_port\":40545,\"dest_ip\":\"192.5.5.241\",\"dest_port\":53,\"proto\":\"UDP\",\"pkt_src\":\"wire/pcap\",\"dns\":{\"version\":2,\"type\":\"answer\",\"id\":24894,\"flags\":\"8010\",\"qr\":true,\"opcode\":0,\"rname\":\"com\",\"rrtype\":\"A\",\"rcode\":\"NOERROR\",\"authorities\":{\"rname\":\"com\",\"rrtype\":\"NS\",\"ttl\":172800,\"rdata\":\"e.gtld-servers.net\"},{\"rname\":\"com\",\"rrtype\":\"NS\",\"ttl\":172800,\"rdata\":\"b.gtld-servers.net\"},{\"rname\":\"com\",\"rrtype\":\"NS\",\"ttl\":172800,\"rdata\":\"a.gtld-servers.net\"},{\"rname\":\"com\",\"rrtype\":\"NS\",\"ttl\":172800,\"rdata\":\"d.gtld-servers.net\"},{\"rname\":\"com\",\"rrtype\":\"NS\",\"ttl\":172800,\"rdata\":\"i.gtld-servers.net\"},{\"rname\":\"com\",\"rrtype\":\"NS\",\"ttl\":172800,\"rdata\":\"f.gtld-servers.net\"},{\"rname\":\"com\",\"rrtype\":\"NS\",\"ttl\":172800,\"rdata\":\"j.gtld-servers.net\"},{\"rname\":\"com\",\"rrtype\":\"NS\",\"ttl\":172800,\"rdata\":\"k.gtld-servers.net\"},{\"rname\":\"com\",\"rrtype\":\"NS\",\"ttl\":172800,\"rdata\":\"c.gtld-servers.net\"},{\"rname\":\"com\",\"rrtype\":\"NS\",\"ttl\":172800,\"rdata\":\"g.gtld-servers.net\"},{\"rname\":\"com\",\"rrtype\":\"NS\",\"ttl\":172800,\"rdata\":\"h.gtld-servers.net\"},{\"rname\":\"com\",\"rrtype\":\"NS\",\"ttl\":172800,\"rdata\":\"l.gtld-servers.net\"},{\"rname\":\"com\",\"rrtype\":\"NS\",\"ttl\":172800,\"rdata\":\"m.gtld-servers.net\"},{\"rname\":\"com\",\"rrtype\":\"DS\",\"ttl\":86400},{\"rname\":\"com\",\"rrtype\":\"RRSIG\",\"ttl\":86400}}}								

Fig: 4.5.6 System logs of ids

4.6 Suricata installation

```
/var/lib/suricata/rules/

# Update system
echo "Updating system..."

sudo apt update && sudo apt upgrade -y

echo "Installing dependencies"

sudo apt install -y software-properties-common wget curl gnupg lsb-release

echo "Adding Suricata PPA"

sudo add-apt-repository -y ppa:oisf/suricata-stable

Update package list again

sudo apt update

Install Suricata

echo "Installing Suricata"

sudo apt install -y suricata

Enable Suricata service

echo "Enabling Suricata service"

sudo systemctl enable suricata

echo "Starting Suricata service"

sudo systemctl start suricata

echo "Verifying Suricata installation"

suricata --build-info

# Update Suricata rules (using Emerging Threats Open Rules)

echo "Updating Suricata rules..."

sudo suricata-update

# Test Suricata

echo "Testing Suricata configuration..."

sudo suricata -T

echo "Suricata installation and setup complete!"
```

4.7 pfsense firewall installation:

```
#!/bin/bash

# pfSense CLI Script Example

# This script should be run on a system with access to pfSense through SSH or console

# Check if running as root
if [[ $EUID -ne 0 ]]; then
    echo "This script must be run as root."
    exit 1
fi

echo "Starting pfSense firewall and network configuration..."

# Variables for network configuration
WAN_INTERFACE="wan" # Adjust based on your setup
LAN_INTERFACE="lan"
LAN_IP="192.168.1.1"
LAN_NETMASK="24"    # CIDR notation
WAN_DHCP=true       # Set to true for DHCP on WAN

# Configure network interfaces
echo "Configuring network interfaces..."

pfSsh.php <<EOF
\${config['interfaces']}[${WAN_INTERFACE}]['enable'] = "true";
\${config['interfaces']}[${WAN_INTERFACE}]['type'] = "dhcp";
\${config['interfaces']}[${LAN_INTERFACE}]['enable'] = "true";
\${config['interfaces']}[${LAN_INTERFACE}]['ipaddr'] = "${LAN_IP}";
\${config['interfaces']}[${LAN_INTERFACE}]['subnet'] = "${LAN_NETMASK}";
write_config();
interface_reconfigure("${WAN_INTERFACE}", true);
interface_reconfigure("${LAN_INTERFACE}", true);
EOF

# Restart network services
```

```

echo "Restarting network services..."

pfctl -d # Disable firewall temporarily during setup

/etc/rc.reload_interfaces

pfctl -e # Re-enable firewall

# Adding basic firewall rules

echo "Adding basic firewall rules..."

pfSsh.php <<EOF

\$rule = array();

\$rule['type'] = "pass";

\$rule['interface'] = "\$LAN_INTERFACE";

\$rule['source']['network'] = "\$LAN_IP/\$LAN_NETMASK";

\$rule['destination']['any'] = true;

\$rule['protocol'] = "tcp/udp";

\$rule['descr'] = "Allow LAN to any";

\$config['filter']['rule'][] = \$rule;

\$rule = array();

\$rule['type'] = "block";

\$rule['interface'] = "\$WAN_INTERFACE";

\$rule['source']['any'] = true;

\$rule['destination']['any'] = true;

\$rule['protocol'] = "icmp";

\$rule['descr'] = "Block ICMP on WAN";

\$config['filter']['rule'][] = \$rule;

write_config();

filter_configure();

EOF

# Verifying and applying rules

echo "Verifying and applying firewall rules..."

pfctl -sr # Display current ruleset

```

```

# Configure NAT (if required)

echo "Configuring NAT for LAN access to WAN..."

pfSsh.php <<EOF

\${nat}_rule = array();

\${nat}_rule['interface'] = "$WAN_INTERFACE";

\${nat}_rule['source']['network'] = "$LAN_IP/$LAN_NETMASK";

\${nat}_rule['protocol'] = "any";

\${nat}_rule['target'] = "";

\${nat}_rule['descr'] = "Outbound NAT for LAN";

\${config['nat']['outbound']['rule'][]} = \${nat}_rule;

\${config['nat']['outbound']['mode']} = "automatic";

write_config();

filter_configure();

EOF

# Final configuration reload

echo "Reloading configuration..."

/etc/rc.filter_configure

echo "pfSense firewall and network configuration complete!"

```

CHAPTER 5

RESULTS AND DISCUSSION

5.1 Results

In the project Building a Virtual Firewall, IPS, and IDS with Suricata, the results demonstrate the effectiveness of integrating Suricata with a pfSense firewall to enhance network security. The system successfully detected and prevented various simulated threats, including DoS attacks and malicious traffic, showcasing the IPS's capability to block offenders through predefined rules. The IDS functionality efficiently logged and alerted administrators to potential security breaches, allowing for real-time monitoring and analysis. By enabling features like "Bypass Firewall Rules for Traffic on the Same Interface," the project ensured flexibility without compromising security. The discussion highlights how rule customization, regular updates (e.g., ET Open Rules), and strategic deployment of Suricata improve overall network defense, offering a robust and cost-effective solution for protecting virtualized environment.

Interfaces	Global Settings	Updates	Alerts	Blocks	Files	Pass Lists	Suppress	Logs View	Logs Mgmt	SID Mgmt
Sync	IP Lists									
Logs Browser Selections										
Instance to View (LAN) LAN										
Choose which instance logs you want to view.										
Log File to View alerts.log										
Choose which log you want to view..										
Status/Result File successfully loaded.										
Log File Path: /var/log/suricata/suricata_vtnet148757/alerts.log										
Refresh										
Log Contents										
08/13/2024:20:36:50.967426 [**] [1:2022973:1] ET INFO Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy										
08/13/2024:21:36:49.163014 [**] [1:2022973:1] ET INFO Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy										
08/13/2024:22:36:47.378518 [**] [1:2022973:1] ET INFO Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy										
08/18/2024:11:04:55.696557 [**] [1:2022973:1] ET INFO Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy										
08/18/2024:12:04:54.975081 [**] [1:2022973:1] ET INFO Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy										
08/18/2024:13:04:55.567061 [**] [1:2022973:1] ET INFO Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy										
08/18/2024:14:04:55.778971 [**] [1:2022973:1] ET INFO Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy										
08/18/2024:15:01:43.259684 [**] [1:2022973:1] ET INFO Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy										
08/18/2024:16:01:35.722125 [**] [1:2022973:1] ET INFO Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy										
08/18/2024:17:00:31.582687 [**] [1:2022973:1] ET INFO Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy										
08/18/2024:20:06:33.438409 [**] [1:2022973:1] ET INFO Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy										
08/18/2024:21:06:34.377446 [**] [1:2022973:1] ET INFO Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy										
08/18/2024:22:06:45.397610 [**] [1:2022973:1] ET INFO Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy										
08/18/2024:23:06:21.940135 [**] [1:2022973:1] ET INFO Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy										
08/19/2024:00:06:50.102502 [**] [1:2022973:1] ET INFO Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy										
08/21/2024:09:52:40.283238 [**] [1:2049335:2] ET FILE_SHARING File Sharing Related Domain in DNS Lookup (send-anywhere .com) [**] [Classification: Misc acti										
08/21/2024:09:52:40.283238 [**] [1:2049335:2] ET FILE_SHARING File Sharing Related Domain in DNS Lookup (send-anywhere .com) [**] [Classification: Misc acti										
08/21/2024:09:52:40.655683 [**] [1:2049363:2] ET FILE_SHARING File Sharing Domain Observed in TLS SNI (send-anywhere .com) [**] [Classification: Misc activi										
08/21/2024:09:52:40.660028 [**] [1:2049363:2] ET FILE_SHARING File Sharing Domain Observed in TLS SNI (send-anywhere .com) [**] [Classification: Misc activi										
08/21/2024:09:52:40.678337 [**] [1:2049363:2] ET FILE_SHARING File Sharing Domain Observed in TLS SNI (send-anywhere .com) [**] [Classification: Misc activi										

.Fig: 5.1 Alerts log

5.2 Discussion

The project underscores the importance of combining IDS and IPS functionalities with a robust firewall to achieve holistic network protection. Suricata's ability to handle both intrusion detection and prevention in a virtualized environment offers a scalable and cost-effective security solution. However, the results also highlight the need for regular rule updates and fine-tuning to minimize false alerts and ensure compatibility with specific network requirements.

The project revealed that while open-source tools like Suricata and pfSense offer extensive capabilities, their effectiveness heavily depends on proper configuration and ongoing maintenance. For instance, updating rulesets and monitoring logs require a disciplined approach to avoid vulnerabilities caused by outdated definitions or misconfigurations. The integration of alerting and reporting tools, such as email notifications or third-party logging systems, could further enhance the system's responsiveness to emerging threats.

In conclusion, the project serves as a valuable case study in implementing a cost-effective and powerful virtual firewall with IDS/IPS capabilities. It provides a strong foundation for further research and development, such as integrating machine learning for anomaly detection or enhancing automation in threat response. The lessons learned and methodologies applied can be extended to other cybersecurity applications, emphasizing the need for continuous evolution to stay ahead of evolving threats.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

The project revealed that while open-source tools like Suricata and pfSense offer extensive capabilities, their effectiveness heavily depends on proper configuration and ongoing maintenance. For instance, updating rulesets and monitoring logs require a disciplined approach to avoid vulnerabilities caused by outdated definitions or misconfigurations. The integration of alerting and reporting tools, such as email notifications or third-party logging systems, could further enhance the system's responsiveness to emerging threats.

In conclusion, the project serves as a valuable case study in implementing a cost-effective and powerful virtual firewall with IDS/IPS capabilities. It provides a strong foundation for further research and development, such as integrating machine learning for anomaly detection or enhancing automation in threat response. The lessons learned and methodologies applied can be extended to other cybersecurity applications, emphasizing the need for continuous evolution to stay ahead of evolving threats.

6.2 Future scope

Future Enhancements:

- 1) Incorporating machine learning-based anomaly detection for better threat identification. Testing the setup in larger, real-world environments to measure scalability.
- 2) Integration with SIEM (Security Information and Event Management) tools for advanced monitoring

REFERENCES

- [1] D. Kuipers and M. Fabro, "Control System Cyber Security: Defense in Depth Strategies," No. INL/EXT-06 11478, 2006
- [2] G. Kumar and S. Panda, "Spectrum of Effective Security Trust Architecture to Manage Interception of Packet Transmission in Value Added Network," *The Global Journal of Computer Science and Technology*, Vol. 11, No. 4, pp. 73-77, 2011.
- [3] B. Caswell, J. Beale, and A. Baker, *Snort IDS and IPS Toolkit*. New York: Syngress, 2007.
- [4] M. Lococo, "Capacity Planning for Snort," retrieved October 2, 2011 from mikelococo.com/2011/08/snort-capacity-planning/.
- [5] T. Weber, "Network Intrusion Detection - Keeping Up with Increasing Information Volume," SANS Institute, 2001.
- [6] P. Mell, V. Hu, R. Lippmann, J. Haines, and M. Zissman, "An Overview of Issues in Testing Intrusion Detection Systems.," retrieved October 4, 2011 from www.net-security.org.
- [7] D. Day and B. Burns, "A Performance Analysis of Snort and Suricata Network Intrusion Detection and Prevention Engines," *Fifth International Conference on Digital Society*, Gosier, Guadeloupe, pp. 187-192, 2011.
- [8] S. Damaye, "Suricata-Vs-Snort," retrieved from www.aldeid.com/wiki/Suricata-vs-snort, October 2, 2011.
- [9] E. Leblond, "Optimizing Linux on Multicore CPUs," retrieved September 18, 2011 home.regit.org/2011/01/optimizing-suricata-on-a-multicore-cpu/. From
- [10] E. Albin, "A Comparative Analysis of the Snort and Suricata Intrusion-Detection Systems," M.S. thesis, U.S. Naval Postgraduate School, September 2011.
- [11] M. Watchinski, "Unusual Snort Performance Stats," retrieved October 2, 2011 comments.gmane.org/gmane.comp.security.ids.snort.general/30527.
- [12] Ramli, M., & Soewito, B. (2023). Monitoring dan Evaluasi Keamanan Jaringan Dengan Pendekatan System Information and Security Management (SIEM). *Faktor Exacta*, 16(1).

- [13] Anugrah, F. T., Ikhwan, S., & AG, J. G. (2022). Implementasi Intrusion Prevention System (IPS) Menggunakan Suricata Untuk Serangan SQL Injection. *Techné: Jurnal Ilmiah Elektroteknika*, 21(2), 199-210. 40
- [14] Mualfah, D., & Riadi, I. (2017). Network forensics for detecting flooding attack on web server. *International Journal of Computer Science and Information Security*, 15(2), 326.
- [15] Widiyari, I. R. (2022). "Siasat" Ukuw (Universitas Kristen Satya Wacana) Website Security Analysis Using Owasu (Open Web Application Security Project). *Jurnal Teknik Informatika (Jutif)*, 3(3), 763-770.
- [16] M. Syani, "Implementasi Intrusion Detection System (IDS) Menggunakan Suricata Pada Linux Debian 9 Berbasis Cloud Virtual Private Servers (VPS)," *Jurnal Infokar*, vol. 1, no. 1, pp. 13-20, 2020.
- [17] B. S. Anggoro and W. Sulisty, "Implementasi Intrusion Prevention System Suricata dengan Anomaly-Based untuk Keamanan Jaringan PT. Grahamedia Informasi," *Seminar Nasional APTIKOM (SEMNASITK)*, pp. 280-288, 201
- [18] Pamungkas, M. H., & Chandra, D. W. (2022). Analisis Pola dan Dampak Serangan Cryptojacking dengan Menggunakan Metode Analisis Dinamis dan Analisis Statis. *JURIKOM (Jurnal Riset Komputer)*, 9(5), 1511-1519.
- [19] Sopaheluwakan, C. R., & Chandra, D. W. (2020). Anti-WebShell PHP Backdoor Scanner pada Linux Server. *ILKOM Jurnal Ilmiah*, 12(2), 143-153.
- [20] Tallane, R. B., & Chandra, D. W. (2022). Implementation Of Intrusion Detection System (Ids) Using Security Onion. *Syntax Literate; Jurnal Ilmiah Indonesia*, 7(10), 14947-14959.
- [21] Aulianita, R., & Martiwi, R. (2021). PENGGUNAAN METODE IDS DALAM IMPLEMENTASI FIREWALL UNTUK PENCEGAHAN SERANGAN Distributed Denial Of Service (DDoS) PADA JARINGAN. *Jusikom: Jurnal Sistem Komputer Musirawas*, 6(2), 94-104.